

 <https://github.com/ColitoBandito/cs4412-project-Collin>

Mapping Mood Profiles and Feature Associations of Spotify Tracks

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Data Mining W01

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KEY EDA INSIGHTS

This initial phase of exploratory analysis has shown a number of patterns that give an idea how to approach the mining phase of this project.

- **Strong Feature Correlations:** A high positive correlation (0.77) was found between Loudness and Energy, while there was a significant negative correlation (-0.71) between acousticness and energy.
- **Data Density:** Early scatter plots showed that Valence and energy had a dense, overlapping distribution across all features. This confirmed that algorithmic clustering is helpful in distinguishing "mood profiles", beyond just regular listener intuition.
- **Feature Skew:** Attributes such as Instrumentalness skewed strongly towards zero, which indicates that, at least within this dataset, the majority of tracks were vocal-centric tracks.

Major Preprocessing Decisions

Data integrity was paramount in ensuring proper behavior of distance-based mining algorithms. Two primary transformations were applied.

- **Standardization:** Some attributes such as Tempo (BPM) and Loudness (db) existed on much larger scales than others, with Valence being only on a scale of (0.0-1.0). As such all numerical attributes were standardized using StandardScaler to eliminate any potential magnitude bias.
- **Dimensionality Reduction:** Principal Component Analysis was utilized to reduce the amount of high dimensional space into two primary components. This was to enable visualization of clusters on a 2D plane without sacrificing maximum variance.

MINING RESULTS AND INTERPRETATION

The Elbow Method was used to determine $k = 4$ as the optimal number of clusters. The rate of decrease in inertia leveled off substantially at this point. Four "Mood Profiles" were discovered. Each Profile is identified by its mathematical center:

Table 1: Discovered Spotify Mood Profiles ($k = 4$)

Cluster	Key Characteristics (Centroids)	Proposed Mood Label
0	High Valence (0.68), High Danceability (0.70)	Upbeat / High Energy Pop
1	High Acousticness (0.84), Low Energy (0.20)	Chill / Acoustic Folk
2	High Energy (0.67), High Loudness (-7.51 dB)	Intense / Modern Rock
3	Highest Speechiness (0.36), High Danceability (0.68)	Vocal-Heavy / Hip-Hop

- Cluster 0 (Upbeat/High Energy): Representative of positive sounding music. High valence suggests the music here contains an optimistic tone, that it's easy dance to and reminiscent of modern Pop.
- Cluster 1 (Chill/Acoustic Folk): The most distinct of all profiles generated, it shows a strong inverse relationship in acousticness and energy. Representative of tracks that are "unplugged" and lower tempo.
- Cluster 2 (Intense / Modern Rock): Energetic but not happy music sonically, this grouping contained music with more serious tones. It includes most tracks where high intensity isn't associated with a positive mood.
- Cluster 3 (Vocal heavy/ Hip Hop): Contains tracks that are on average 5x more wordy than the other groups. Groups music that is high in wordiness, which encompasses rap/hip hop that is performed on top of instrumentation.

PLANS FOR MILESTONE 3

The aim for Milestone 3 is to move away from general track groupings and move towards identifying specific feature relationships while detecting all anomalous track profiles. In accordance with that aim, this next phase will utilize Association Rule Mining using the Apriori Algorithm. This will be facilitated by specifying numerical features into categories ("High", "Medium", "Low"). This ought to reveal non-obvious patterns. An example I foresee as one of these patterns are tracks with high instrumentalness within specific tempo ranges. I will utilize DB Scan to identify "Anomalous Track Profiles".